

## **Unit-1: Statistics & Data Science Notes**

### **1. Types of Data**

**Data is information collected for analysis. It can be classified into different types.**

#### **A. Categorical Data (Qualitative)**

**Categorical data represents labels or groups.**

**Types**

##### **i) Nominal Data**

**No order exists.**

**Categories are only names.**

**Examples**

- Gender: Male, Female
- Blood Group: A, B, AB, O
- Colors: Red, Blue, Green

Example Table

##### **Student Blood Group**

|   |   |
|---|---|
| A | O |
| B | A |
| C | B |

No blood group is greater than another.

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##### **ii) Ordinal Data**

Categories have an order.

Examples

- Education Level
  - High School
  - Bachelor's
  - Master's
  - PhD
- Customer Satisfaction
  - Poor
  - Fair
  - Good
  - Excellent

Although "Excellent" is higher than "Good", the difference between them is not measurable.

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#### **B. Numerical Data (Quantitative)**

Numbers representing measurable quantities.

##### **i) Discrete Data**

Countable values.

Examples

- Number of students
- Number of cars
- Number of books

Example

Students in class = 45

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## ii) Continuous Data

Can take any value within a range.

Examples

- Height = 172.4 cm
  - Weight = 63.75 kg
  - Temperature = 27.8°C
- 

## Summary

| Type       | Examples           |
|------------|--------------------|
| Nominal    | Gender, Color      |
| Ordinal    | Rank, Satisfaction |
| Discrete   | Number of students |
| Continuous | Height, Weight     |

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## 2. Descriptive Statistics

Descriptive statistics summarize data.

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### A. Measures of Central Tendency

They describe the center of data.

#### Mean

Average value.

Formula

$$\text{Mean} = \sum X / n$$

Example

Marks

20, 30, 40, 50, 60

Mean

$$= 20 + 30 + 40 + 50 + 60 / 5 = 40$$


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#### Median

Middle value after sorting.

Example

2, 4, 5, 8, 10

Median = 5

Even number example

2, 4, 6, 8

Median

$$= 4 + 6 / 2 = 5$$

Median is useful when outliers exist.

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## **Mode**

Most frequent value.

Example

2,3,3,4,5

Mode = 3

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## **Comparison**

### **Measure Best Used When**

Mean     Normal data

Median   Outliers present

Mode     Categorical data

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## **B. Measures of Dispersion**

Shows spread of data.

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### **Range**

Largest – Smallest

Example

10,20,30,40

Range

40–10=30

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### **Variance**

Measures average squared distance from mean.

Population Formula

$$\sigma^2 = \sum (x - \mu)^2 / N$$

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### **Standard Deviation**

Square root of variance.

Small SD → Data close together.

Large SD → Data widely spread.

Example

Class A

50,51,49,50

Very small SD

Class B

20,40,60,80

Large SD

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## **C. Skewness**

Measures asymmetry.

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### **Symmetric Distribution**

Mean = Median = Mode

Bell-shaped curve.

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### **Positive Skew (Right Skew)**

Long tail on right.

Mean > Median > Mode

Example

Salary distribution  
Many people earn average salary.  
Few earn extremely high salary.

---

### **Negative Skew (Left Skew)**

Long tail on left.  
Mean < Median < Mode  
Example  
Easy exam  
Most students score high.  
Few score low.

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### **D. Kurtosis**

Measures peak and tail heaviness.

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#### **Mesokurtic**

Normal distribution.  
Kurtosis = 3

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#### **Leptokurtic**

Very sharp peak.  
Heavy tails.  
Many outliers.

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#### **Platykurtic**

Flat peak.  
Light tails.  
Few outliers.

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### **3. Data Visualization**

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#### **Histogram**

Shows frequency distribution.

Example

Exam Scores

0-10 \*\*\*

10-20 \*\*\*\*\*

20-30 \*\*\*\*\*

30-40 \*\*\*\*

Used for continuous data.

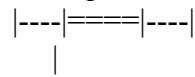
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#### **Box Plot**

Shows

- Minimum
- Q1
- Median
- Q3
- Maximum
- Outliers

Example



Median

Useful for detecting outliers.

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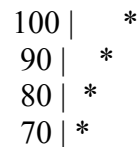
### Scatter Plot

Shows relationship between two variables.

Example

Hours Studied vs Marks

Marks



Study Hours

Positive trend.

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### Pair Plot

Shows scatter plots for every pair of variables.

Useful in machine learning.

Example

Variables

- Age
- Salary
- Height

Pair plot creates

Age vs Salary

Age vs Height

Salary vs Height

and distributions.

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## 4. Probability Distributions

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### A. Normal Distribution

Bell-shaped.

Properties

- Mean = Median = Mode
- Symmetric
- Continuous

Examples

- IQ
- Height
- Measurement errors

68–95–99.7 Rule

Within

- 1 SD → 68%
- 2 SD → 95%
- 3 SD → 99.7%

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## **B. Binomial Distribution**

Number of successes.

Conditions

- Fixed trials
- Two outcomes
- Constant probability

Example

10 coin tosses.

Probability of getting 7 heads.

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## **C. Poisson Distribution**

Counts events.

Example

- Emails received per hour
- Accidents per day
- Calls arriving

Parameter

$\lambda$  (average count)

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## **D. Exponential Distribution**

Time between events.

Examples

Time until

- Next customer arrives
- Next phone call
- Next machine failure

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## **E. Uniform Distribution**

All values equally likely.

Example

Rolling a fair die.

Each number

Probability =  $1/6$

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## **F. Beta Distribution**

Values between 0 and 1.

Used for

- Probabilities
- Bayesian statistics

Example

Probability customer buys a product.

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### **G. Gamma Distribution**

Positive continuous values.

Examples

- Waiting times
  - Insurance claims
  - Rainfall amount
- 

### **Distribution Summary**

| <b>Distribution</b> | <b>Used For</b>          |
|---------------------|--------------------------|
| Normal              | Heights                  |
| Binomial            | Success/Failure          |
| Poisson             | Event Counts             |
| Exponential         | Waiting Time             |
| Uniform             | Equal Probability        |
| Beta                | Probabilities            |
| Gamma               | Positive Continuous Data |

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### **5. Covariance**

Measures how two variables change together.

Positive

Both increase together.

Negative

One increases while another decreases.

Example

Study Hours ↑

Marks ↑

Positive covariance.

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### **6. Correlation**

Measures strength of relationship.

Formula

$$-1 \leq r \leq 1$$

Values

$$r = 1$$

Perfect positive.

$$r = -1$$

Perfect negative.

$$r = 0$$

No relationship.

Example  
Hours studied vs marks  
 $r = 0.9$   
Strong positive.

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### Correlation vs Covariance

| Covariance        | Correlation       |
|-------------------|-------------------|
| Depends on units  | Unit-free         |
| Hard to interpret | Easy to interpret |

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### 7. Confounding Variable

A third variable affecting both variables.

Example

Ice cream sales  $\uparrow$

Drowning cases  $\uparrow$

Hidden variable

Summer temperature.

Ice cream does NOT cause drowning.

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### 8. Correlation Matrix

Table showing correlation among variables.

Example

|            | Age | Salary | Experience |
|------------|-----|--------|------------|
| Age        | 1   | 0.8    | 0.9        |
| Salary     | 0.8 | 1      | 0.75       |
| Experience | 0.9 | 0.75   | 1          |

Diagonal always = 1.

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### 9. Distance Measures

Used in clustering, KNN, recommendation systems.

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#### Euclidean Distance

Straight-line distance.

Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example

A = (2, 3)

B = (5, 7)

Distance

$$= \sqrt{9 + 16} = 5$$

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#### Manhattan Distance

City-block distance.

Formula

$$|x_2 - x_1| + |y_2 - y_1|$$

Example

(2, 3)

(5, 7)

Distance

$$3+4=7$$

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### Cosine Distance

Measures angle between vectors.

Formula

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$
$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

Example

Used in

- Text similarity
- Recommendation systems
- NLP

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### Distance Comparison

| Distance | Best Use |
|----------|----------|
|----------|----------|

|           |                 |
|-----------|-----------------|
| Euclidean | Continuous data |
|-----------|-----------------|

|           |            |
|-----------|------------|
| Manhattan | Grid paths |
|-----------|------------|

|        |                 |
|--------|-----------------|
| Cosine | Text similarity |
|--------|-----------------|

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## 10. Sampling Techniques

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### Simple Random Sampling

Every member has equal chance.

Example

Randomly selecting 100 students from 1000.

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### Stratified Sampling

Divide population into groups.

Sample from every group.

Example

Students

- 60% Male
- 40% Female

Maintain same ratio.

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### Cluster Sampling

Divide into clusters.

Choose entire clusters.

Example

Schools

Randomly choose 5 schools.

Survey every student.

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### Comparison

| Sampling      | Used When                        |
|---------------|----------------------------------|
| Simple Random | Equal selection chance           |
| Stratified    | Represent all groups             |
| Cluster       | Population spread geographically |

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### 11. Central Limit Theorem (CLT)

The sampling distribution of the sample mean approaches a normal distribution as the sample size becomes large (commonly  $n \geq 30$ ), regardless of the population's original distribution, provided the observations are independent and the population has finite variance.

Example

Population income is skewed.

Take many samples of 50 people.

Calculate each sample mean.

The distribution of these sample means becomes approximately normal.

Importance

- Confidence intervals
  - Hypothesis testing
  - Statistical inference
- 

### 12. Law of Large Numbers (LLN)

As sample size increases, the sample average approaches the true population average.

Example

Coin toss

10 tosses

Heads = 8

Probability = 0.80

10,000 tosses

Heads  $\approx 5,000$

Probability  $\approx 0.50$

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### CLT vs LLN

| CLT                                      | LLN                                      |
|------------------------------------------|------------------------------------------|
| Sample means become normally distributed | Sample mean converges to population mean |
| Used for inference                       | Used for convergence                     |

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### 13. Data Preprocessing

Data preprocessing prepares raw data for analysis and machine learning.

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#### A. Handling Missing Values

Methods

- Delete rows with missing values (when few records are missing)
- Delete columns with too many missing values
- Fill with Mean (for numerical data)
- Fill with Median (robust to outliers)

- Fill with Mode (for categorical data)
- Use advanced methods such as KNN imputation or regression imputation for better estimates

Example

**Age**

20

22

NA

25

Mean =  $(20 + 22 + 25) / 3 = 22.33$

Replace NA with 22.33 (or use median = 22 if preferred).

## B. Outlier Treatment

Outliers are unusually large or small values.

Detection Methods

- Box plot
- Z-score (commonly  $|Z| > 3$ )
- IQR Rule

IQR Method

$IQR = Q3 - Q1$

Outlier if

- Below  $Q1 - 1.5 \times IQR$
- Above  $Q3 + 1.5 \times IQR$

Treatment

- Remove outliers (if due to errors)
- Cap/Winsorize extreme values
- Transform data (e.g., logarithm)
- Use robust models

## C. Encoding Categorical Variables

Algorithms require numerical inputs.

### Label Encoding

Assign integers to categories.

Example

#### Color Encoded

Red 0

Blue 1

Green 2

Suitable for ordinal categories.

### One-Hot Encoding

Create binary columns.

Example

### Color Red Blue Green

Red 1 0 0

Blue 0 1 0

Green 0 0 1

Suitable for nominal categories.

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### D. Normalization and Standardization

Feature scaling ensures variables with different units are comparable.

#### Min–Max Normalization

Scales values to the range [0, 1].

Example

Data = 10, 20, 30

Value = 20

Normalized value =  $(20 - 10) / (30 - 10) = 0.5$

#### Standardization (Z-score Scaling)

Centers data around mean 0 with standard deviation 1.

$$z = \frac{x - \mu}{\sigma} \quad z = \frac{x - \mu}{\sigma}$$

Example

Mean = 50, SD = 10

Value = 70

Z-score =  $(70 - 50) / 10 = 2$

Normalization is commonly used for neural networks, while standardization is widely used for algorithms such as linear regression, logistic regression, support vector machines, and principal component analysis.

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## END

## C01

**Summarize and interpret datasets using descriptive measures, sampling and distributional analysis.**

P0/PS0 Level

Justification

P01 3 Applies mathematics, probability, and statistics fundamentals for data analysis.

P02 2 Identifies data characteristics and performs exploratory analysis.

| P0/PS0 Level |   | Justification                                                     |
|--------------|---|-------------------------------------------------------------------|
| P04          | 2 | Conducts statistical investigations using descriptive techniques. |
| P05          | 2 | Uses statistical software and programming tools for analysis.     |
| P011         | 1 | Encourages independent learning of statistical methods.           |
| PS01         | 2 | Forms the foundation for AI and Data Science applications.        |
| PS02         | 2 | Uses programming tools such as Python/R for statistical analysis. |

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## C02

**Perform statistical inference using estimation, hypothesis testing, and resampling approaches.**

| P0/PS0 Level |   | Justification                                                    |
|--------------|---|------------------------------------------------------------------|
| P01          | 3 | Applies statistical theory and mathematical concepts.            |
| P02          | 3 | Analyzes data to draw valid statistical conclusions.             |
| P04          | 3 | Conducts hypothesis testing and interprets experimental results. |
| P05          | 2 | Uses statistical software for inference.                         |
| P011         | 2 | Learns modern statistical inference techniques.                  |
| PS01         | 3 | Essential for developing AI and ML models.                       |
| PS02         | 2 | Implements inference using modern programming tools.             |

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## C03

**Build, evaluate, and interpret regression and regularized models.**

| P0/PS0 Level |   | Justification                                               |
|--------------|---|-------------------------------------------------------------|
| P01          | 3 | Applies mathematical concepts in regression modeling.       |
| P02          | 3 | Analyzes relationships among variables.                     |
| P03          | 2 | Develops predictive models for engineering applications.    |
| P04          | 3 | Investigates model performance using experimental analysis. |
| P05          | 3 | Uses ML libraries and software tools.                       |
| P011         | 2 | Learns advanced regression techniques.                      |
| PS01         | 3 | Core competency in Machine Learning.                        |
| PS02         | 3 | Implements models using modern AI frameworks.               |

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## C04

**Apply supervised and unsupervised learning algorithms effectively.**

| P0/PS0 Level |   | Justification                                |
|--------------|---|----------------------------------------------|
| P01          | 3 | Applies engineering and computing knowledge. |

| P0/PS0 Level |   | Justification                                            |
|--------------|---|----------------------------------------------------------|
| P02          | 3 | Selects appropriate ML algorithms for problem solving.   |
| P03          | 3 | Designs intelligent learning solutions.                  |
| P04          | 2 | Evaluates learning algorithms experimentally.            |
| P05          | 3 | Uses AI frameworks and programming tools effectively.    |
| P06          | 1 | Considers societal implications of intelligent systems.  |
| P08          | 1 | Supports collaborative implementation of ML projects.    |
| P011         | 2 | Encourages learning emerging AI techniques.              |
| PS01         | 3 | Directly addresses AI and Machine Learning development.  |
| PS02         | 3 | Uses modern AI libraries and frameworks.                 |
| PS03         | 2 | Applies AI solutions considering responsible deployment. |

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## C05

### Evaluate models using appropriate metrics, Bayesian reasoning, and cross-validation.

| P0/PS0 Level |   | Justification                                               |
|--------------|---|-------------------------------------------------------------|
| P01          | 2 | Applies statistical and mathematical evaluation techniques. |
| P02          | 3 | Critically analyzes model performance.                      |
| P03          | 2 | Improves solution quality through evaluation.               |
| P04          | 3 | Performs investigations using validation techniques.        |
| P05          | 3 | Uses ML evaluation tools and software.                      |
| P07          | 1 | Encourages responsible and unbiased model evaluation.       |
| P09          | 1 | Communicates evaluation findings effectively.               |
| P011         | 2 | Adopts modern evaluation methodologies.                     |
| PS01         | 3 | Essential for AI model development.                         |
| PS02         | 3 | Uses modern ML frameworks for evaluation.                   |
| PS03         | 2 | Evaluates fairness and responsible AI performance.          |

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## C06

### Analyze fairness, model interpretability, and ethical implications in ML workflows.

| P0/PS0 Level |   | Justification                                                |
|--------------|---|--------------------------------------------------------------|
| P01          | 1 | Applies basic engineering knowledge to responsible AI.       |
| P02          | 2 | Analyzes ethical and interpretability issues.                |
| P03          | 2 | Designs transparent and explainable AI systems.              |
| P04          | 2 | Investigates fairness and bias using analytical methods.     |
| P05          | 2 | Uses interpretability and fairness assessment tools.         |
| P06          | 3 | Evaluates societal and environmental impacts of AI.          |
| P07          | 3 | Strongly aligns with professional ethics and responsible AI. |

| P0/PS0 Level |   | Justification                                                                    |
|--------------|---|----------------------------------------------------------------------------------|
| P08          | 1 | Encourages teamwork in ethical AI development.                                   |
| P09          | 1 | Communicates ethical implications and model transparency.                        |
| P011         | 2 | Promotes continuous learning in trustworthy AI.                                  |
| PS01         | 2 | Supports development of responsible AI systems.                                  |
| PS02         | 2 | Uses modern explainable AI and fairness tools.                                   |
| PS03         | 3 | Directly focuses on ethical, socially responsible, and sustainable AI solutions. |

### Summary

- **P01, P02, P04, and P05** receive the strongest mapping because the course is grounded in statistics, machine learning, experimentation, and modern computational tools.
- **P03** is moderately addressed through predictive model development and AI solution design.
- **P06 and P07** are primarily addressed through the course outcome on fairness, interpretability, and ethics.
- **PS01 and PS02** have strong alignment across most COs since the course develops AI/ML knowledge and practical implementation skills.
- **PS03** is strongly achieved through **CO6**, with moderate support from CO4 and CO5 where responsible AI practices and fair model evaluation are introduced.

## Chap-2

### 1. Point Estimation

#### Definition

A **point estimate** is a single numerical value used to estimate an unknown population parameter.

- Population Mean ( $\mu$ )  $\rightarrow$  Sample Mean ( $\bar{x}$ )
- Population Variance ( $\sigma^2$ )  $\rightarrow$  Sample Variance ( $s^2$ )
- Population Proportion ( $p$ )  $\rightarrow$  Sample Proportion ( $\hat{p}$ )

#### Example

Suppose marks of five students are  
20, 25, 30, 35, 40

Sample Mean

$$\bar{x} = 20 + 25 + 30 + 35 + 40 / 5 = 30$$

Hence,

Point estimate of population mean = **30**

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#### Types of Estimators

| Population Parameter    | Estimator                       |
|-------------------------|---------------------------------|
| Mean ( $\mu$ )          | Sample Mean ( $\bar{x}$ )       |
| Variance ( $\sigma^2$ ) | Sample Variance ( $s^2$ )       |
| Proportion ( $p$ )      | Sample Proportion ( $\hat{p}$ ) |

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### 2. Properties of Estimators

A good estimator should satisfy the following properties.

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## A. Unbiasedness

An estimator is **unbiased** if its expected value equals the true parameter.

$$E(\hat{\theta}) = \theta \quad E(\hat{\theta}) = \theta$$

where

- $\theta$  = Population parameter
- $\hat{\theta}$  = Estimator

### Example

Suppose population mean

$$\mu = 50$$

Repeated samples give

49, 51, 50, 50

Average estimate

$$\frac{49+51+50+50}{4} = 50$$

Therefore

Estimator is unbiased.

---

## B. Bias

Bias measures systematic error.

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

Example

True mean = 100

Estimator average = 103

Bias

$$103 - 100 = 3$$

Estimator overestimates by 3.

---

## C. Variance

Variance measures variability of estimator.

$$\text{Var}(\hat{\theta})$$

Lower variance means estimates remain close together.

### Example

Estimator A

98, 99, 100, 101, 102

Estimator B

80, 90, 100, 110, 120

Estimator A has smaller variance.

---

## D. Mean Square Error (MSE)

Combines bias and variance.

$$\text{MSE} = \text{Variance} + \text{Bias}^2$$

Formula

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

Example

Variance = 4

Bias = 2

Then

$$MSE = 4 + 2^2 = 8$$

## 3. Maximum Likelihood Estimation (MLE)

### Definition

MLE estimates parameters that maximize the probability of observing the given data.

Likelihood Function

$$L(\theta) = P(\text{Data} | \theta)$$

Maximum likelihood estimate

$$\hat{\theta} = \arg\max_{\theta} L(\theta)$$

### Steps

1. Write likelihood function.
2. Take logarithm.
3. Differentiate.
4. Set derivative equal to zero.
5. Solve parameter.

### Example

Suppose observations

3, 5, 7

Estimate mean  $\mu$  of Normal distribution.

MLE for mean

$$\hat{\mu} = \bar{x} = \frac{3+5+7}{3} = 5$$

Hence

MLE estimate = 5

### Applications

- Machine Learning
- Logistic Regression
- Naïve Bayes
- Neural Networks

## 4. Method of Moments

### Definition

Method of Moments estimates parameters by equating  
Sample moments = Population moments

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### First Moment

Population

$$E(X) = \mu \quad E(X) = \mu$$

Sample

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Set

$$\mu = \bar{x} \quad \mu = \bar{x}$$

---

### Example

Sample

2, 4, 6

Mean

$$\bar{x} = 4 \quad \bar{x} = 4$$

Estimated parameter

$$\mu = 4 \quad \mu = 4$$

---

## Difference Between MLE and Method of Moments

| MLE                       | Method of Moments   |
|---------------------------|---------------------|
| Uses likelihood           | Uses moments        |
| More accurate             | Easier              |
| Computationally intensive | Simple calculations |

---

## 5. Confidence Intervals

A confidence interval gives a range within which the population parameter is expected to lie.

General Formula

$$\text{Estimate} \pm \text{Margin of Error} \quad \text{Estimate} \pm \text{Margin of Error}$$

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## A. Confidence Interval for Mean

When  $\sigma$  is known

$$\bar{x} \pm z \frac{\sigma}{\sqrt{n}} \quad \bar{x} \pm z \frac{\sigma}{\sqrt{n}}$$

Example

Mean=70

$\sigma=10$

$n=100$

95% confidence

$z=1.96$

Margin

$$1.96 \cdot 1010 = 1.961.96 \cdot \frac{10}{10} = 1.961.96 \cdot 1010 = 1.96$$

CI

$$70 \pm 1.9670 \pm 1.96$$

$$=(68.04, 71.96)$$

---

## B. Confidence Interval for Proportion

Formula

$$\hat{p} \pm z_{\hat{p}(1-\hat{p})n} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} \pm z_{\hat{p}(1-\hat{p})}$$

Example

Sample proportion

0.60

n=200

95%

Margin

$$1.96 \cdot 0.6(0.4)2001.96 \sqrt{\frac{0.6(0.4)}{200}}1.962000.6(0.4)$$

$$\approx 0.068$$

CI

$$0.532 \text{ to } 0.668$$

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## C. Confidence Interval for Variance

Formula

$$(n-1)s^2 \times \frac{1}{\chi^2_{\alpha/2}} \times \frac{1}{\chi^2_{1-\alpha/2}}$$

Uses Chi-square distribution.

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## 6. Hypothesis Testing

### Steps

1. State hypotheses
  2. Choose significance level  $\alpha$
  3. Calculate statistic
  4. Compare with critical value/p-value
  5. Decision
- 

### Null Hypothesis

$H_0$

No difference.

Example

Average salary = ₹50,000

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## Alternative Hypothesis

$H_1$  Difference exists.

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## 7. Z-Test

Used when

- Large sample ( $n > 30$ )
- Population variance known

Formula

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \quad z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Example

$\mu = 100$

Sample mean = 105

$\sigma = 20$

$n = 64$

$$z = \frac{105 - 100}{20 / \sqrt{64}} = \frac{5}{2.5} = 2$$

---

## 8. t-Test

Used when

- Small sample
- Variance unknown

Formula

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} \quad t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Example

$n = 10$

Mean = 52

Hypothesized mean = 50

$s = 4$

$$t = \frac{52 - 50}{4 / \sqrt{10}} = \frac{2}{1.26} = 1.58$$

---

## 9. Chi-Square Test

Used for

- Independence
- Goodness of fit
- Variance testing

Formula

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \chi^2 = \sum \frac{(O - E)^2}{E}$$

Example

Observed

40

Expected

50

Contribution

$$(40-50) \frac{250}{50} = -2 \frac{(40-50)^2}{50} = -2 \frac{100}{50} = -2$$

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